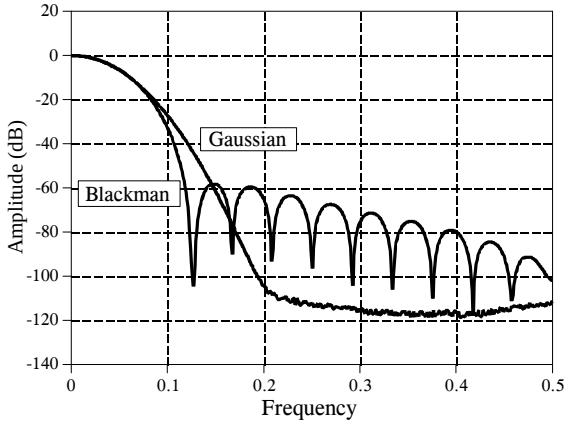

FIGURE 15-4
Frequency response of the Blackman window and Gaussian filter kernels. Both these filters provide better stopband attenuation than the moving average filter. This has no advantage in removing random noise from time domain encoded signals, but it can be useful in mixed domain problems. The disadvantage of these filters is that they must use *convolution*, a terribly slow algorithm.

algorithm, imagine passing an input signal, $x[\ ]$, through a seven point moving average filter to form an output signal, $y[\ ]$. Now look at how two adjacent output points, $y[50]$ and $y[51]$, are calculated:

$$y[50] = x[47] + x[48] + x[49] + x[50] + x[51] + x[52] + x[53]$$

$$y[51] = x[48] + x[49] + x[50] + x[51] + x[52] + x[53] + x[54]$$

These are nearly the same calculation; points $x[48]$ through $x[53]$ must be added for $y[50]$, and again for $y[51]$. If $y[50]$ has already been calculated, the most *efficient* way to calculate $y[51]$ is:

$$y[51] = y[50] + x[54] - x[47]$$

Once $y[51]$ has been found using $y[50]$, then $y[52]$ can be calculated from sample $y[51]$, and so on. After the first point is calculated in $y[\ ]$, all of the other points can be found with only a single addition and subtraction per point. This can be expressed in the equation:

EQUATION 15-3
Recursive implementation of the moving average filter. In this equation, $x[\ ]$ is the input signal, $y[\ ]$ is the output signal, M is the number of points in the moving average (an odd number). Before this equation can be used, the first point in the signal must be calculated using a standard summation.

$$y[i] = y[i-1] + x[i+p] - x[i-q]$$

where: $p = (M-1)/2$

$q = p + 1$

Notice that this equation use two sources of data to calculate each point in the output: points from the input *and* previously calculated points from the output. This is called a **recursive** equation, meaning that the result of one calculation